xAI Reading Group: Week 1

The Two Cultures in Statistical Modeling and Why Explainable AI

Rakhi Singh

Outline

1. Introduction
2. Primer: The Two Cultures of Modeling (Breiman, 2001)
3. What is Interpretability?
4. Why Interpretability?
5. Interpretability vs. Explainability
6. Goals of Explainability
7. References

Dynamics of the group

A weekly reading group in the JAN-MAY 2026 semester.

We will meet on Thursdays from 3:30 to 5 pm in KCB 504.

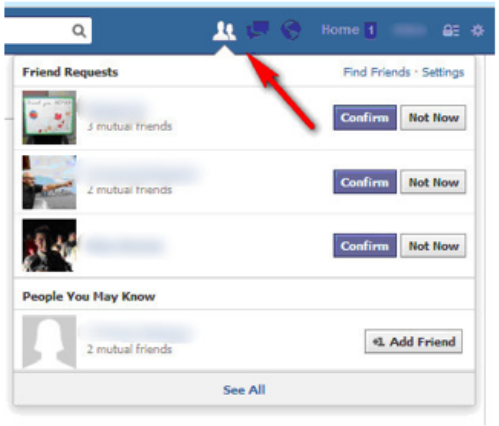
While no prior research experience is required, since you signed up for this, you should have an interest and willingness to think deeply.

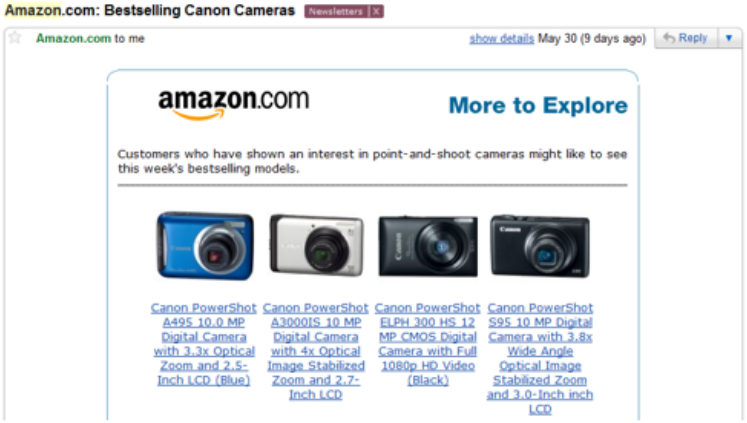
Each week, we will read one research paper (or chapter/article). The aim is to achieve a shared understanding, not a formal evaluation.

Responsibilities will rotate between

- **paper leader** (person who will give a 15-minute overview),
- **discussants** (everyone brings 1 insight + 1 question),
- **scribe** posts a short summary after the session.

Machine Learning is Everywhere!





Source: `https://interpretable-ml-class.github.io/`

Statistical Modeling: Two Cultures

Leo Breiman (2001) describes two dominant cultures in data modeling:

- **Data Modeling Culture**:
  - focuses on specifying probabilistic models for data, interpreting parameters, and testing hypotheses.
  - Typical tools: linear regression, generalized linear models, and parametric statistics.
  - Primary goal: **Understanding the data-generating process**
- **Algorithmic Modeling Culture**
  - focuses on treating the data mechanism as unknown, learn patterns directly from data, and optimizing predictive accuracy.
  - Typical tools: random forests, neural networks, boosting, SVMs.
  - Primary goal: **Prediction, not explanation**

These cultures differ in goals, assumptions, and methods.

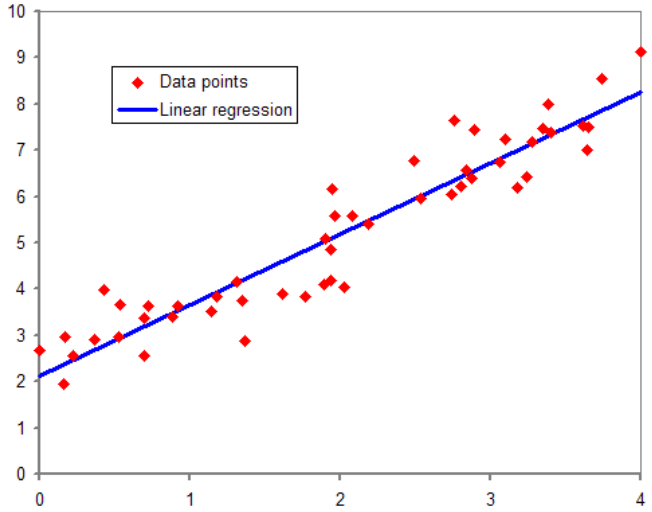
Prediction vs Explanation Example

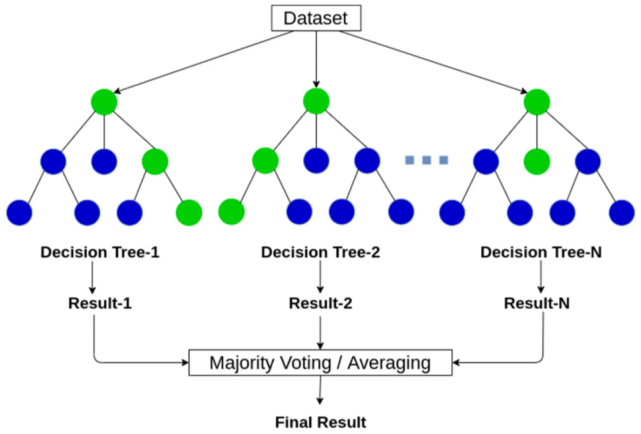
Data Culture:

$$Price = 100 + 50 \cdot Bedrooms + 20 \cdot Location + \epsilon$$

Algorithmic Culture:

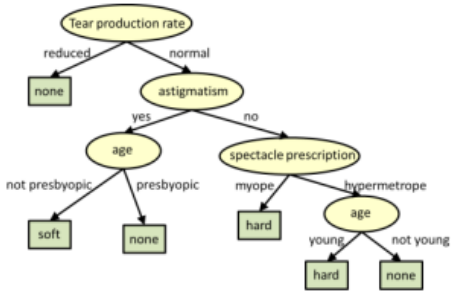
- Random Forest
- 500 decision trees
- Hard to summarize



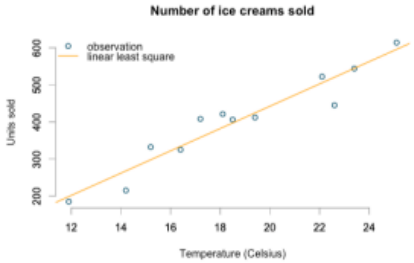


More examples

Interpretable Models

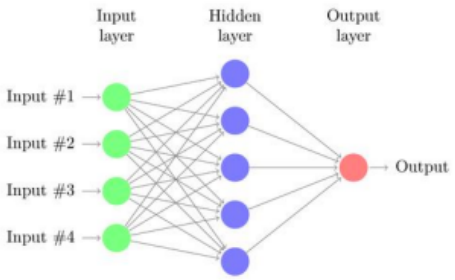


Decision Trees

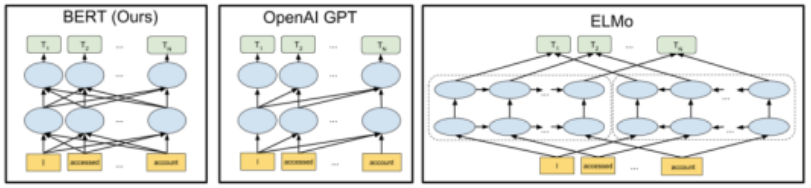


Linear Models

Black Boxes



Simple Neural Networks



Complex Neural Networks

Breiman's Critique and The Modern Landscape

Breiman argued that statistics overemphasized simple, interpretable models, strong assumptions, inference over prediction.

He advocated for complex models, fewer assumptions, empirical performance.

Today, algorithmic models dominate: deep learning, large-scale deployment, automated decision systems.

But this success introduces new problems such as opacity, bias, lack of accountability,and loss of scientific insight.

High-stakes domains require accurate predictions, understandable reasoning, and accountable decisions.

Neither culture alone is sufficient.

From Two Cultures to Interpretability

Interpretability emerges as a bridge between:

- the predictive power of algorithmic models,
- and the understanding sought by data modeling.

Goal:

Retain accuracy while enabling human understanding.

Why not just use simple models?

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Why not just use simple models? They are often insufficiently accurate, unable to capture complex structure, and too rigid for modern data.

Why not use complex models? They are often powerful, but opaque, and difficult to trust.

Interpretability addresses this trade-off.

What is Interpretability?

Interpretability is not just a model property.

It depends on the user (developer, doctor, citizen), the task (debugging, auditing, decision support), and the context (risk and consequences).

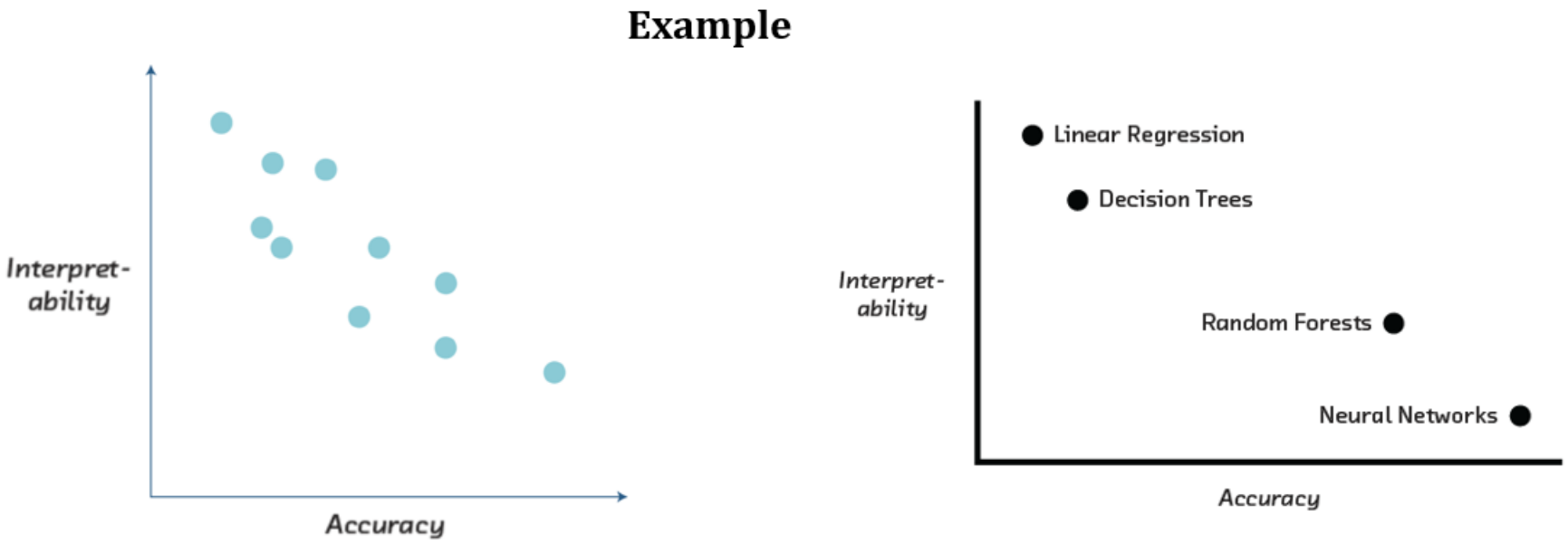
An explanation for an engineer may fail for a patient.

What is it? There is no single universal definition.
Interpretability depends on:

- who is interpreting,
- what they need to know,
- how complex the model is.

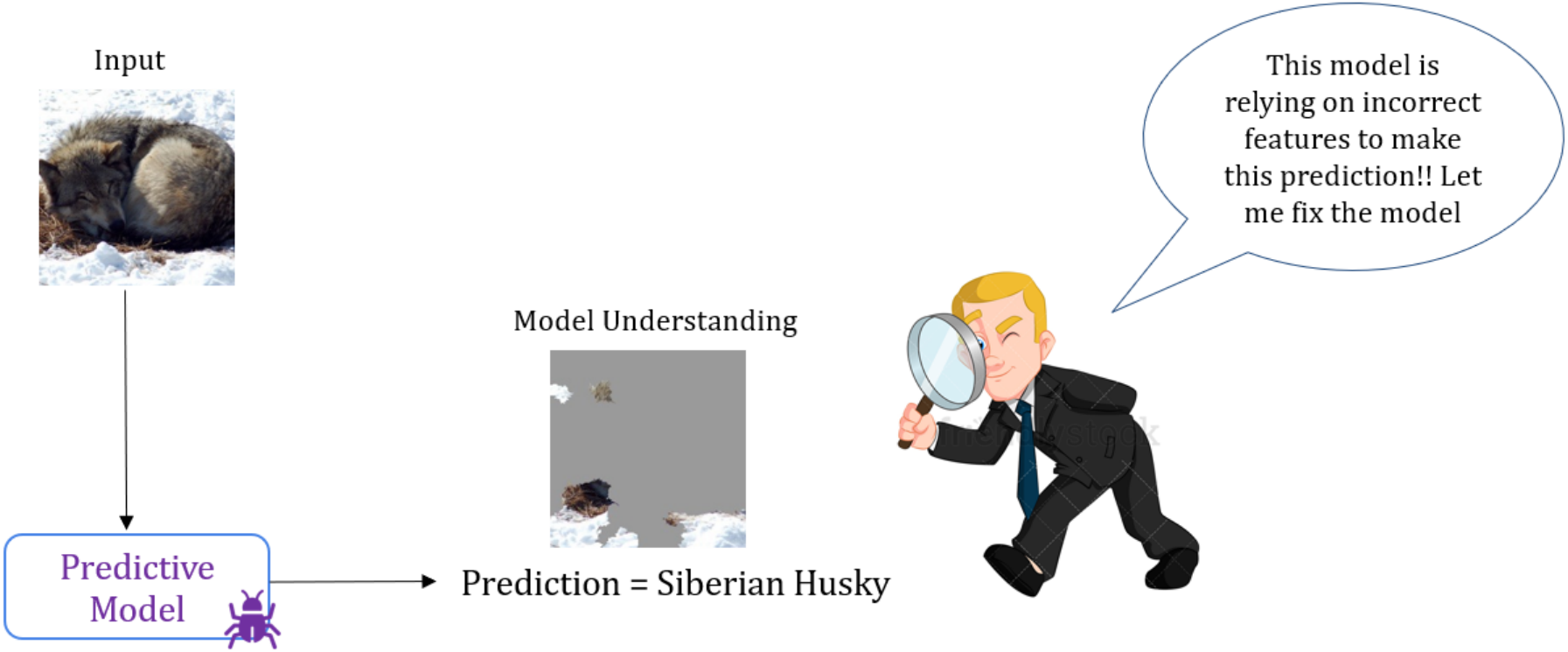
A model is interpretable if a human can understand its behavior for a given purpose.

Interpretability vs. Accuracy



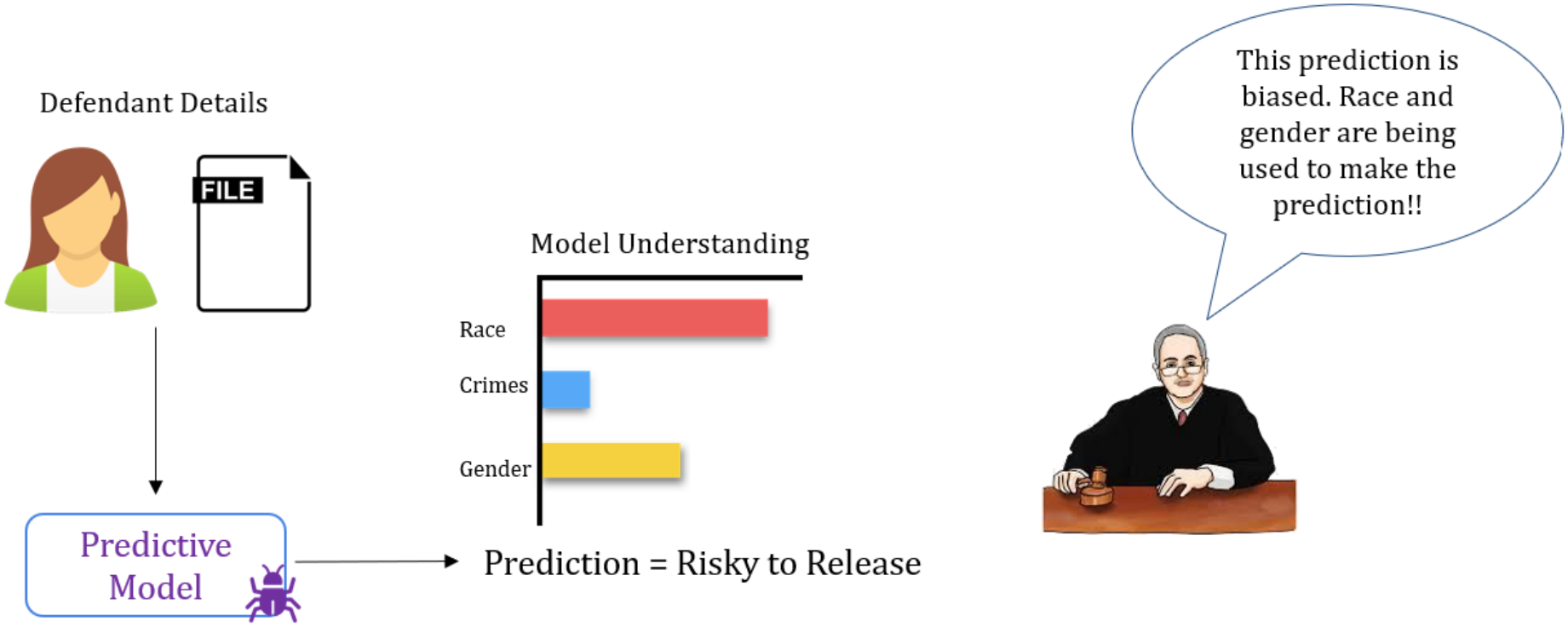
Source: `https://interpretable-ml-class.github.io/`

Why: Model understanding facilitates debugging



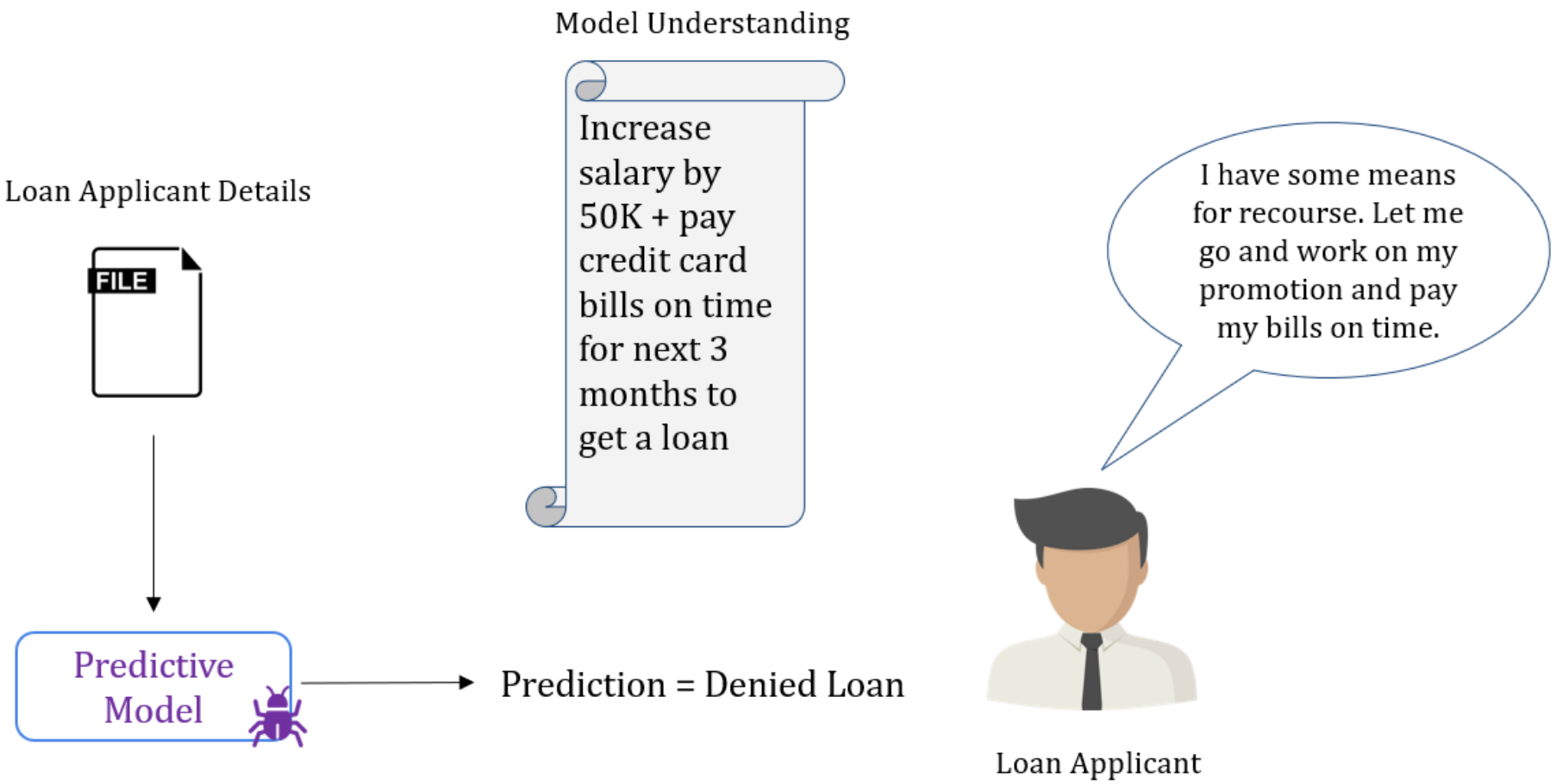
Source: `https://interpretable-ml-class.github.io/`

Why: Model understanding facilitates bias detection



Source: `https://interpretable-ml-class.github.io/`

Why: Model understanding facilitates recourse for adversely affected



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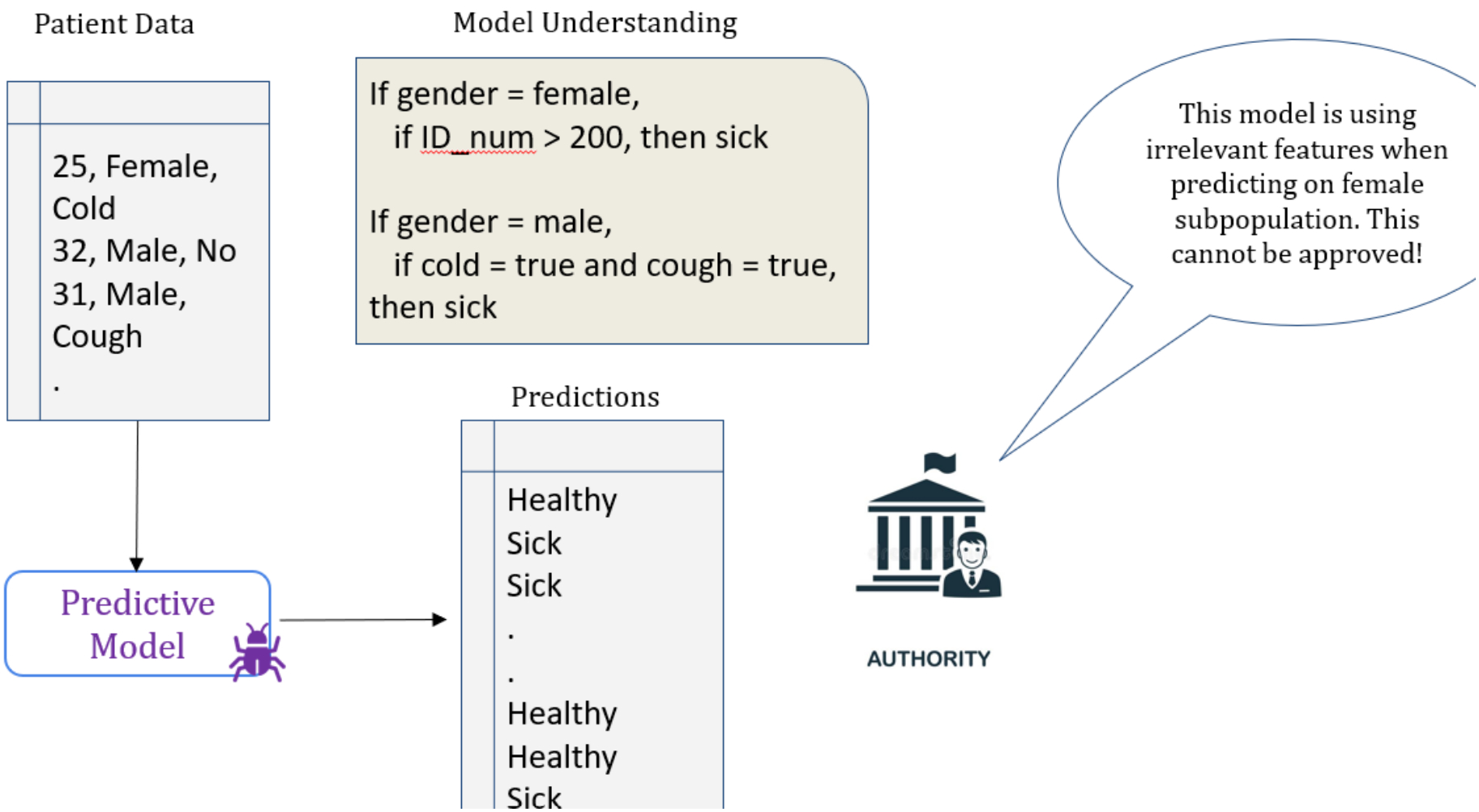
Why: Model understanding helps assess when to trust



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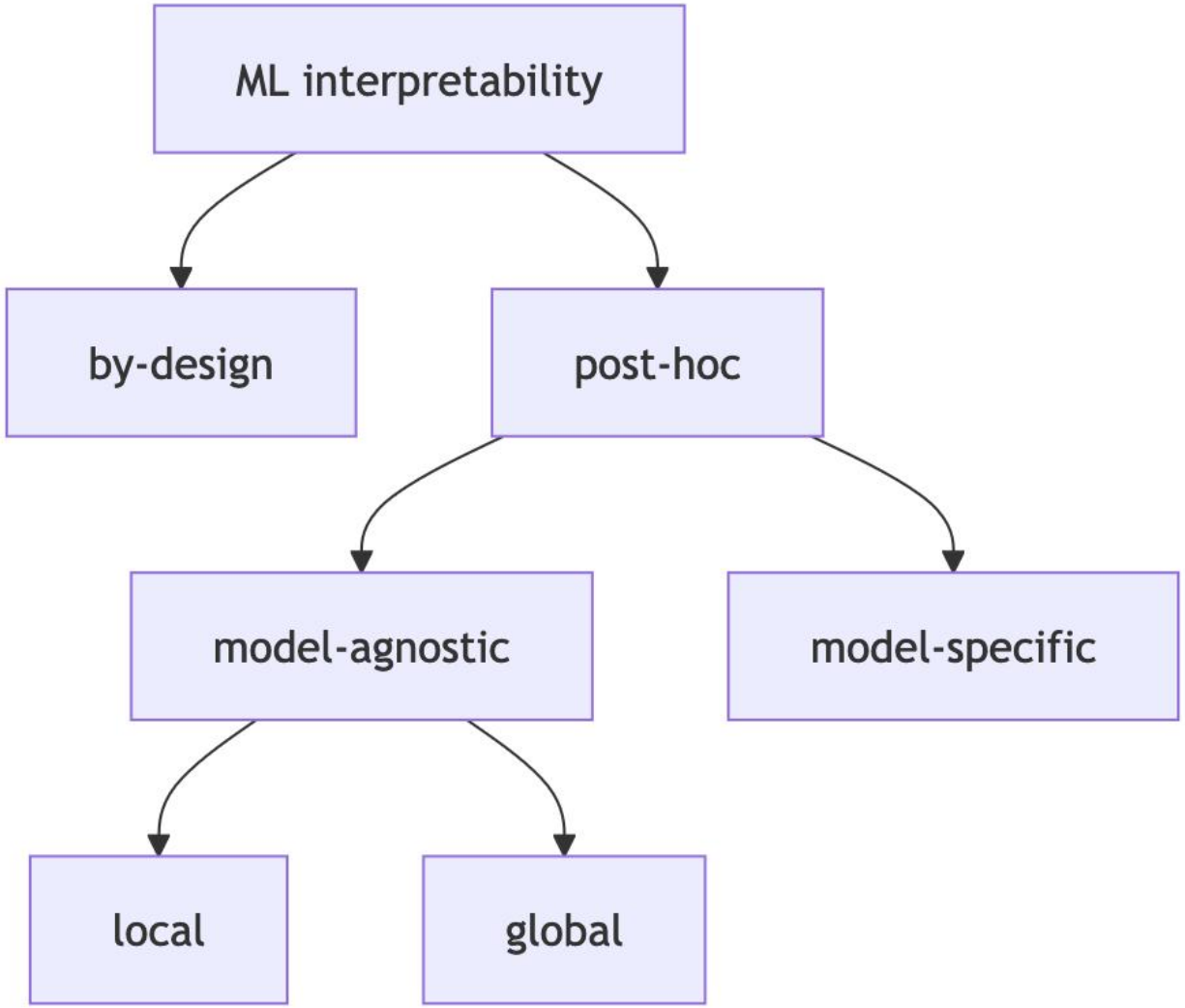
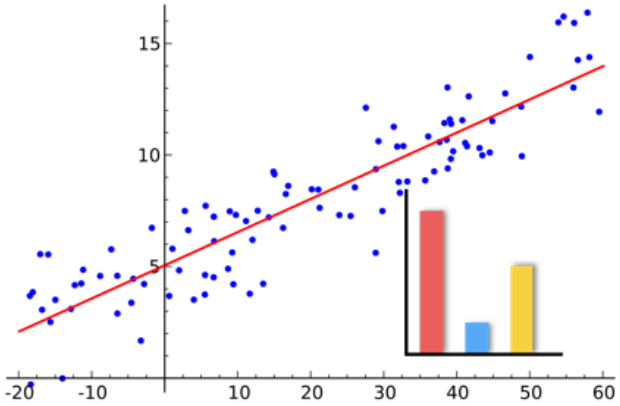
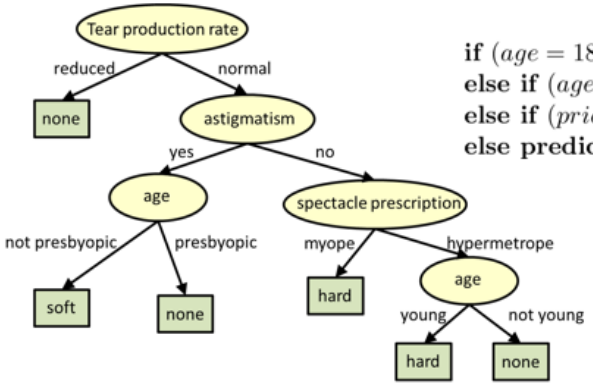
Types of interpretability/explainability



Take 1: Build *inherently interpretable* predictive models

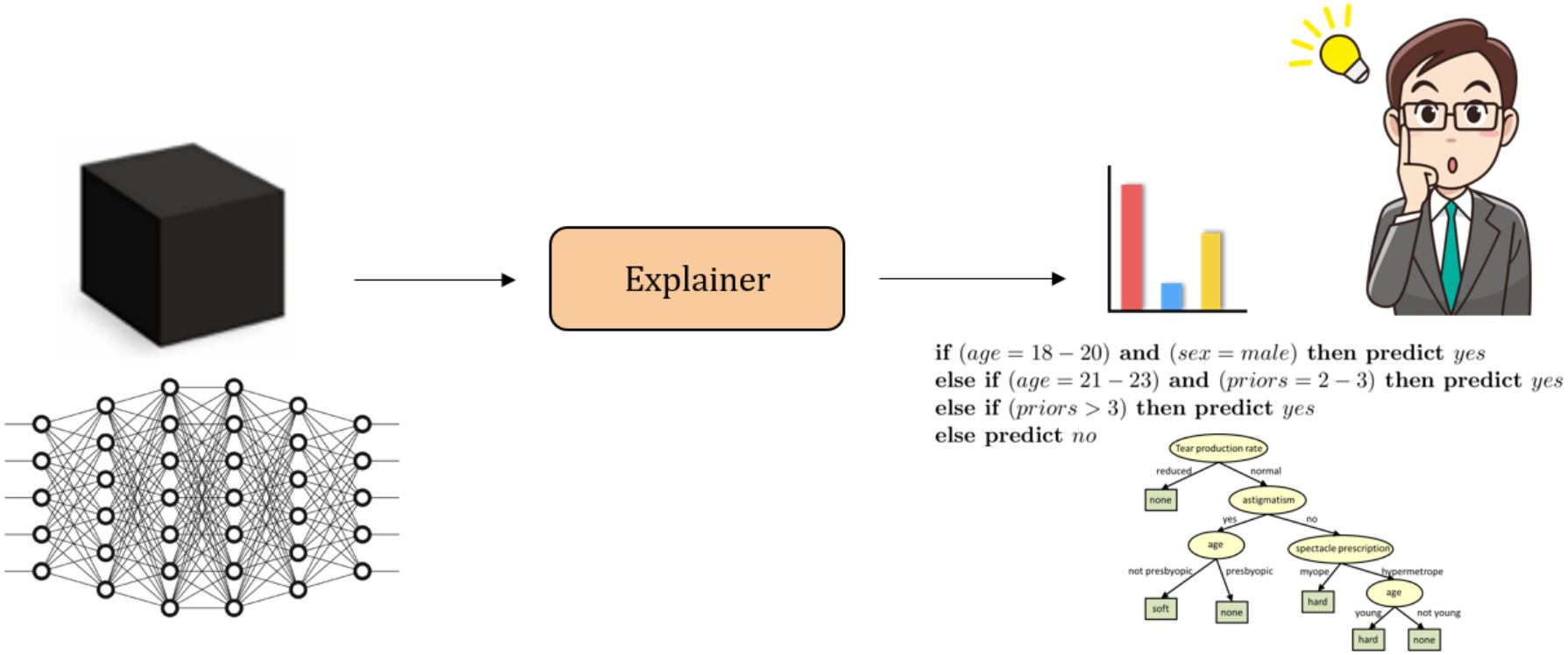




if $(age = 18 - 20)$ **and** $(sex = male)$ **then predict** yes
else if $(age = 21 - 23)$ **and** $(priors = 2 - 3)$ **then predict** yes
else if $(priors > 3)$ **then predict** yes
else predict no

Source: `https://interpretable-ml-class.github.io/`

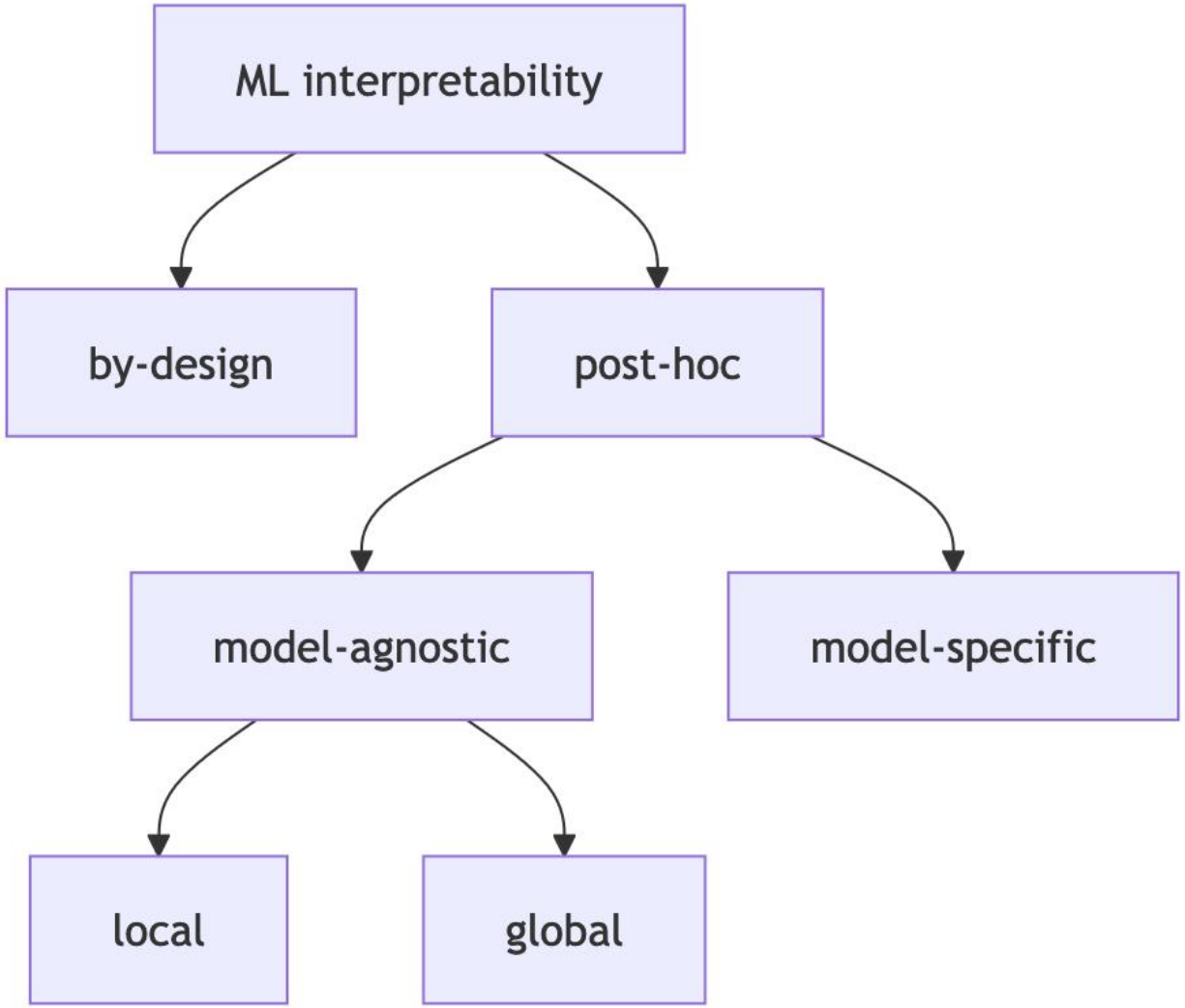
Take 2: *Explain* pre-built models in a *post-hoc manner*



Explanations are approximations, not true reasoning.

Source: `https://interpretable-ml-class.github.io/`

Types of interpretability/explainability



https://christophm.github.io/interpretable-ml-book/overview.html

Types of Interpretability / Explainability (contd)

Model-specific

- exploit internal structure
- e.g., Decision Trees, Linear Models

Model-agnostic

- treat model as black box
- e.g., LIME, SHAP

Global interpretability

- overall model behavior
- feature importance, PDP

Local interpretability

- individual predictions
- instance-level explanations

Representative Methods

- Decision Trees
- Linear Models
- Feature Importance
- Partial Dependence Plots (PDP)
- LIME
- SHAP

(Examples will be shown later)

What Can Explainability Do?

1. **Increase Trust**
   - Understand uncertainty
   - Compare model mistakes with human mistakes
   - Decide when automation is acceptable
2. **Reveal (Limits of) Causality**
   - What can models tell us about the world?
   - Models learn correlations, not causes
   - Decisions require more than prediction
3. **Improve the model**
   - Detect spurious correlations and shortcuts
   - Reveal unintended model behavior
   - Enable debugging and robustness (Clever Hans)

Clever Hans

Clever Hans (German: *der Kluge Hans*; c. 1894 – c. 1916) was a horse that appeared to perform arithmetic and other intellectual tasks during exhibitions in Germany in the early 20th century.

Hans was a horse owned by Wilhelm von Osten, who was a gymnasium mathematics teacher, an amateur horse trainer and phrenologist and was considered to be a mystic.[1] Hans was said to have been taught to add, subtract, multiply, divide, work with fractions, tell the time, keep track of the calendar, differentiate between musical tones, and read, spell, and understand German. Von Osten would ask Hans, "If the eighth day of the month comes on a Tuesday, what is the date of the following Friday?" Hans would answer by tapping his hoof eleven times. Questions could be asked both orally and in written form. Von Osten exhibited Hans throughout Germany and never charged admission. Hans' abilities were reported in *The New York Times* in 1904.[3]

Discussion

- Is interpretability a scientific necessity or a social requirement?
  - Scientific: to understand mechanisms and generate knowledge
  - Social: to ensure fairness, transparency, and accountability
- Can explanations ever be faithful, or are they always approximations?
  - Faithful: simple models (linear regression, small trees)
  - Approximate: SHAP, LIME explain behavior, not true internal reasoning
- Should we sacrifice accuracy for interpretability in high-stakes settings?
  - Yes: healthcare, criminal justice, credit decisions
  - No: weather prediction, recommendation systems
- Inherently interpretable models vs. post-hoc explanations. Which do you trust more and why?
  - Interpretable models: transparent but sometimes less accurate
  - Post-hoc explanations: flexible but potentially misleading

Extra references

- Interpretable Machine Learning
- Explainable AI by Pushpak Bhattacharya
- Harvard's course
- Towards Rigorous Interpretability
- DSEG-LIME: Improving Image Explanation by Hierarchical Data-Driven Segmentation
- XRAI: Better Attributions Through Regions
- Shapley explainability on the data manifold
- Understanding Global Feature Contributions With Additive Importance Measures
- Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead
- Interpretability Beyond Feature Attribution: Quantitative Testing with Concept Activation Vectors (TCAV)
- Questioning the AI: Informing Design Practices for Explainable AI User Experiences
- Explainable machine-learning predictions for the prevention of hypoxaemia during surgery